

Installation

Hardware

RPI 4 - 4GB (8GB Recommended)

Flash Card

RPI Camera V2

Arduino UNO Rev3 or compatible board

Your Sentry Gun Device

Software

PSG 2021 files including Arduino sketch From Github and forcenet

<https://github.com/Darksplat/Fire-Control-Software>

<https://www.forcenet.gov.au/community/viewgroup?Id=d265cfc8-44b6-4c5c-bfac-4ac0ad57b9f0>

These files require their extensions and file names changed before use.

PSG 2021/Arduino Files/PSG_2020/PSG_2020.txt => .ino

PSG 2021/static/psgcass.txt => psg.css

PSG 2021/static/psgjs.txt => msg.js

PSG 2021/templates/Web Page => The contents of this file needs to be made into a web document. The easiest way is to open a text editor like notepad++ or Visual Studio Code or Textedit, and save it with the extension .HTML. So the file should be Index.html in the 'templates' folder

PSG 2021/detection.txt => .ini

PSG 2021/piserialports.txt => .py

PSG 2021/ports.txt => .py

PSG 2021/psgini.txt => .ini

PSG 2021/psgpy.txt => .py

Raspberry Pi imager

Raspberry OS - 'Buster' - This is because 'Bullseye' uses a different library for the Plcam and as such will break the install. See this webpage on how to install 'Buster'

<https://www.raspberrypi.com/news/new-old-functionality-with-raspberry-pi-os-legacy/>

On the machine to which the Arduino is connected, download Python 3.9.9 not Python 3.10 as it doesn't have the dependancies that are required at the time of writing. (06/12/2021). From python.org, and install.

This webpage shows how on multiple system types ie MacOS, Windows, Linux

<https://wsvincent.com/install-python/>

MacOS Instructions

It should install `pip`, the package installer. In a cmd/terminal window, run:(copy and paste this code)

```
python3 pip install --upgrade pip
```

This could take a while depending on your internet connection

```
Sudo pip3 install Flask imutils opencv-contrib-python opencv-python pyserial
```

That installs the packages upon which PSG depends.

Download the files from Github, Forcenet

Windows Instructions

It should install `pip`, the package installer. In a cmd/terminal window, run:(copy and paste this code)

```
python3 pip install --upgrade pip
```

This could take a while depending on your internet connection

```
Sudo pip3 install Flask imutils opencv-contrib-python opencv-python pyserial
```

That installs the packages upon which PSG depends.

Download the files from Github, Forcenet

Unzip into wherever you want it to run on the same machine.

Raspberry Pi/Linux/Unix Instructions

It should install `pip`, the package installer. In a cmd/terminal window, run:(copy and paste this code)

```
python3 pip install --upgrade pip
```

This could take a while depending on your internet connection

```
Sudo pip3 install Flask imutils opencv-contrib-python opencv-python pyserial
```

That installs the packages upon which PSG depends.

Download the files from Github, Forcenet

Configuration

Under the unzip'd `psg-2021` folder, you should find `psg.ini`:

To find usb devices on Mac OS open terminal and type this command

```
ls /dev/tty.usb*
```

Type the usb location number into the red number next to com port below ie #com port:

[Web Server]

Host = localhost

Port = 8080

(MacOS)

[Arduino]

COM Port = **/dev/tty.usbmodem.....**

Baud rate = 9600

(Windows)

[Arduino]

COM Port = **COM7**

Baud rate = 9600

(RPI/Linux/Unix)

[Arduino]

COM Port = **/dev/ttyACMO.....**

Baud rate = 9600

[Controller]

Command frequency (Hz): **2**

Uncomment the “Arduino” section, and set “COM Port” to whatever the Arduino is connected to. I’d suggest starting with “COM3”.

The baud rate should be correct (9600 symbols/second). But can be increased if you wish

The command frequency under “Controller” defines how often the PSG program sends the current set of parameters to the Arduino. If it doesn’t send it frequently, it looks the Arduino goes to sleep, returning the turret to the default position. I’ve set it to 2 (twice per second), though perhaps change it to 0.5 (once every two seconds).

In theory (!), you should now be able to run the program; from a cmd window, simply run:

```
python3 psg.py
```

Then open your browser and type this in the address bar localhost:8080

Operation

Firstly, it’s ugly. Secondly, functionality is very limited. This is a “minimum viable product”, as I went to test a few assumptions to make sure I’m on the right track.

Point a web browser at the machine (or from the machine) on which the Arduino is running, at port 8080. You should see something like this:

“pan” field value until you’re happy it’s as far left as you want to go. Then click [Set Pan (left)].

2. Repeat the procedure, tilting the turret to the right. When happy, click [Set Pan (right)].
3. Now we adjust the “tilt” value until it’s as far down as required. Click [Set Tilt (down)].
4. Finally, adjust “tilt” to its maximum required value, and click [Set Tile (up)].
5. Now we calibrate the screen coordinates. Enter 90, 90 into the “pan” and “tilt”, and click [Move]. Once the turret has stopped moving, click [Fire]. Where you see it hit the screen in the view, left-click on that spot. Then click [Set] next to “Centre”.
6. Now, move the turret to the top-left (ideally with the same minimum pan and maximum tilt as noted earlier), and again [Fire]. Click where it hits, and click [Set] beside “Northwest”.
7. Click the [Calibrate] button, and you’re good to go!

Active use

Click on the “Active” radio button. You should now be able to click on the screen, and the pan/tilt values should be calculated so as to hit that point on the screen. Click [Move] to get the turret to get there, and then you can [Fire] at will.

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